

Практикум D4

104 marks of the archive, six techniques, eleven tasks. Everything the archive asks about a discrete random variable that is not binomial, from May 2021 to November 2025: tables with letters in them, means, variances, a variable built from another, and the two Paper 3 investigations that derive the rules themselves.

О практикуме

Собран из трёх тем: 50 баллов `statistics.discrete_random_variables` (0.562 — 50 из 89), 48 `statistics.expected_value` (0.800 — 48 из 60) и 11 производящей функции из `statistics.probability` (0.046 — 11 из 237). 109 баллов по корпусу, 104 рабочих: пять — ноябрьский дубль 2023 года. Первая тема статистики, стоящая на трёх бумагах почти поровну: 41 балл Paper 1, 30 Paper 2, 33 Paper 3.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d4-discrete

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D4 — Discrete random variables and expectation

104 marks of the archive, six techniques, eleven tasks. Everything the archive asks about a discrete random variable that is *not* binomial, from May 2021 to November 2025: tables with letters in them, means, variances, a variable built from another, and the two Paper 3 investigations that derive the rules themselves.

This is the first statistics topic spread over all three papers: about two fifths of the marks are on Paper 1, a third on Paper 3.

The one idea

A discrete random variable is a **table**: the values it can take, and the probability of each.

A table is a distribution only if every probability is between 0 and 1, and all of them together add up to 1

.

That sentence is the whole of the first technique. When a table has a letter in it, those two rules are the equations and inequalities for the letter — the question does not need to state them, and usually does not. And a letter that solves the equation but makes one cell negative is not an answer: the markscheme gives a separate mark for saying so.

And the idea that carries the rest

Everything else about X is a **weighted sum over the table**:

$$\begin{aligned} E(X) &= \sum x \\ &\quad P(X = x), \\ \text{Var}(X) &= \sum (x - E(X))^2 P(X = x) = E(X^2) - E(X)^2 \end{aligned}$$

A frequency table is a table too, with probabilities $f / \sum f$. A generating

function is the table again, written as a polynomial.

How the checks work

They do not know the answers. Each check is handed **the table and the conditions** the question gives:

```
X = Dist({0: 0.41, 1: k - 0.28, 2: 0.46, 3:
0.29 - 2*k**2})
verify_letters('2b', 0.3, k, [X])
```

is *"your k: does it make this table a distribution?"* The check solves the table's own rules itself, throws away the roots that break a cell, and compares. `Expect(X)` and `Var(X)` add up over the table; `P(X < Y)` adds up over pairs of values.

When you are wrong the check says **how**:

what you wrote	what the check says
a root that makes a cell negative	which cell, and what it equals
$0 < q < 1$ for a probability	the ends belong: a probability can be 0 or 1
a range from one cell alone	the cells also add up to one
$E(X^2)$ for the variance	the square of the mean has not been subtracted
$b \text{Var}(T)$ for $\text{Var}(a - bT)$	the multiplier has not been squared
the largest probability for the mode	the mode is the value that has it
$G(t)$ with coefficients reversed	the opposite count

A number built from a letter you rounded on the way — $p = 0.398$, $b = 20$ — is **accepted, with a remark**, as the markscheme accepts it.

Order of work

level	what it means	tasks
	one letter in a table, and the mean	1-3
	two letters and two conditions; spread and a variable built from X	4-9
	Paper 3: no last value, and the table as a polynomial	10- 11

Every task is a real past-paper question, cited.

63 of these 104 marks are on a calculator paper, and the number flatters the calculator. On Paper 3 the generating function is fractions and the mean of the first-success variable is $1/p$. Real GDC work is about a dozen marks: a cubic, and a system from a mean and a variance.

```

import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit.py
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Dist, Freq, Geo, P(), Expect, Var, Pgf

language('en')                 # this
notebook is in English, and so are the checks

# Dist({value: probability}) is a table;
letters are allowed inside.
# Freq({value: frequency}) is a frequency
table read as a distribution.
# Comparisons are events, P() adds them up;
Expect and Var add up over the table.
#      X = Dist({1: 0.2, 2: 0.5, 3: 0.3})
#      P(X >= 2), Expect(X), Var(5 - 2*X)
# The letter E is taken by e = 2.718..., so
the mean is Expect(X).

print('ready; sympy', sp.__version__)
X = Dist({1: 0.2, 2: 0.5, 3: 0.3})
print('the table:          ', X)
print('P(X >= 2):          ', sympify(P(X >=
2)))
print('E(X), Var(X):      ',
sympify(Expect(X)), sympify(Var(X)))
print('G(t):              ', Pgf(X, t))

```

Map of the six techniques

#	technique	you recognise it by	it reduces to
1	a table must be a distribution	a letter in the table, <i>find k</i> , <i>range of r</i>	sum = 1, every cell in $[0, 1]$
2	the mean	$E(X)$, <i>mean</i> , <i>expected number</i>	$\sum x P(X = x)$
3	two letters, two conditions	two unknowns and one more fact: $E(X)$, a variance, $P(X < Y)$	a system
4	spread, and a variable built from X	Var, $2 - X$, <i>multiplied by 10</i> , $a - bt$	$E(X^2) - E(X)^2$, $a^2 \text{Var}(X)$
5	a first success	<i>until the first</i> : values 1, 2, 3, ... with no end	a series
6	a generating function	$G(t) = \sum P(X = x)t^x$	the table as a polynomial

Techniques 1 and 2 are read straight off the table. Technique 3 is technique 1 with one equation too few, and the question supplies it. Technique 4 is technique 2 applied to X^2 or to $aX + b$.

Techniques 5 and 6 are Paper 3: the same sums, where the table has no end or has been turned into a polynomial.

Part 1. The table

Theory: the two rules, and a letter inside

A game spinner scores X points, with

x	0	1	2	3
$P(X = x)$	0.25	k^2	$1 - 1.6k$	$k - 0.2$

Rule two gives the equation. The four cells add to one:

$$\begin{aligned}0.25 + k^2 + 1 - 1.6k + k - 0.2 &= 1 \\&\implies k^2 - 0.6k + 0.05 \\&= 0 \\&\implies k \\&= 0.1 \text{ or } k \\&= 0.5\end{aligned}$$

Rule one decides between the roots.

Put each back into *every* cell:

- $k = 0.1$ makes $P(X = 3) = 0.1 - 0.2 = -0.1$. Not a probability — rejected;
- $k = 0.5$ gives 0.25, 0.25, 0.2, 0.3. All in $[0, 1]$, sum 1.

Write the reason. " $k = 0.1$ is rejected because $P(X = 3) < 0$ " is a mark of its

own (R1). The number 0.5 without it is not the full answer.

A range instead of a value. If a table is $m, 3m, 1 - 4m$ for $x = 1, 2, 3$, the sum is already one for every m , and only rule one is left: $m \geq 0, 3m \leq 1, 1 - 4m \geq 0$, so

$$0 \leq m \leq \frac{1}{4}$$

Both ends belong: a probability *can* be 0. And the range comes from **all** the cells together — $3m \leq 1$ alone would allow $m = \frac{1}{3}$, where the last cell is $-\frac{1}{3}$.

The mode is the value with the largest probability — for the spinner, $x = 3$. Not 0.3: that is its probability.

Task 1 ● — **May 2025 TZ3 Paper 2**
Q3(a)-(b), 5 marks

A supermarket analyses the shopping habits of its customers. The number of times, X , each customer visits the supermarket in a week is given by the following probability distribution.

x	1	2	3	4	5	≥ 6
$P(X = x)$	$1.5a$	$2a$	0.281	a	0.026	0

(a) (i) Find the value of a . **(ii)** Write down the mode of X .

(b) Find the mean of X .

The mean check does not use your a : it finds a from the table itself, so a slip in (a)(i) costs you (a)(i) only.

```
qla = ...          # a
qlm = ...          # the mode of X
qlb = ...          # the mean of X

a = symbols('a')
visits = Dist({1: 1.5*a, 2: 2*a, 3: 0.281, 4:
a, 5: 0.026, 6: 0}) # 6 stands for "6 or
more"

verify_letters('la(i)', qla, a, [visits])
verify_mode('la(ii)', qlm, visits, given=[],
var=a)
verify_moment('lb', qlb, Expect(visits),
given=[], var=a)
```

Theory: the mean is a weighted sum

For the spinner above, with $k = 0.5$:

$$\begin{aligned}
 E(X) &= 0 \times 0.25 \\
 &\quad + 1 \times 0.25 \\
 &\quad + 2 \times 0.2 \\
 &\quad + 3 \times 0.3 \\
 &= 1.55
 \end{aligned}$$

Every value is multiplied by **its own** probability. $\frac{0+1+2+3}{4} = 1.5$ is the average of the values — the mean of a spinner that lands on each value equally often, which this one does not.

$E(X)$ need not be a value X can take, and need not be a whole number: it is where the table balances.

A frequency table is a distribution. A football team's goals in 20 matches:

goals	0	1	2	3
matches	4	7	6	3

Read each frequency as a probability by dividing by 20. The estimated probability of scoring in a match is $\frac{7+6+3}{20} = 0.8$, and the expected number of goals is

$$\frac{0 \times 4 + 1 \times 7 + 2 \times 6 + 3 \times 3}{20} = \frac{28}{20} = 1.4$$

Over a season of 38 matches that is $38 \times 1.4 = 53.2$ goals expected — the mean scales with the number of matches.

"Expected" is not "will happen". 53.2 goals is a fine expected value. But *"how many buses must run"* asks for a whole number, and it is rounded **up**: a bus for part of the demand still has to run.

Task 2 ● — **May 2022 TZ1 Paper 2 Q3, 6 marks**

A discrete random variable, X , has the following probability distribution:

x	0	1	2	3
$P(X = x)$	0.41	$k - 0.28$	0.46	$0.29 - 2k^2$

- (a) Show that $2k^2 - k + 0.12 = 0$.
- (b) Find the value of k , giving a reason for your answer.
- (c) Hence, find $E(X)$.

For (a), write the equation itself, as *Eq(left, right)*: any rearrangement of it is accepted.

```

k = symbols('k')

q2a = ...          # the equation for k
q2b = ...          # k
q2c = ...          # E(X)

X = Dist({0: 0.41, 1: k - 0.28, 2: 0.46, 3:
0.29 - 2*k**2})

verify_equation('2a', q2a,
Eq(total_probability(X), 1), var=k)
verify_letters('2b', q2b, k, [X])
verify_moment('2c', q2c, Expect(X), given=[],
var=k)

```

Task 3 — May 2023 TZ1 Paper 1 Q2, 6 marks

On a Monday at an amusement park, a sample of 40 visitors was randomly selected as they were leaving the park. They were asked how many times that day they had been on a ride called The Dragon. This information is summarized in the following frequency table.

number of times on The Dragon	0	1	2	3	4
frequency	6	16	13	2	3

It can be assumed that this sample is representative of all visitors to the park

for the following day.

(a) For the following day, Tuesday, estimate **(i)** the probability that a randomly selected visitor will ride The Dragon; **(ii)** the expected number of times a visitor will ride The Dragon.

It is known that 1000 visitors will attend the amusement park on Tuesday. The Dragon can carry a maximum of 10 people each time it runs.

(b) Estimate the minimum number of times The Dragon must run to satisfy demand.

```
q3p = ...          # P(a visitor rides The
Dragon)
q3e = ...          # the expected number of
rides per visitor
q3r = ...          # the minimum number of runs

rides = Freq({0: 6, 1: 16, 2: 13, 3: 2, 4:
3})

verify_chance('3a(i)', q3p, P(rides >= 1))
verify_moment('3a(ii)', q3e, Expect(rides))
verify_moment('3b', q3r, Expect(rides * 1000
/ 10))
```

Part 2. Two letters, and ranges

Theory: one equation is not enough for two letters

A table with two unknowns,

x	0	1	2	3
$P(X = x)$	a	b	0.3	0.1

gives only $a + b = 0.6$ from its sum. The question has to add a fact — say $E(X) = 1.05$ — and that fact is the second equation:

$$\begin{array}{rcl} a & & \\ & + b & = 0.6 \\ 0 \cdot a & & \\ & + 1 \cdot b & \\ & + 2(0.3) & \\ & + 3(0.1) & = 1.05 \\ \implies & & \\ b = 0.15, & & \\ a = 0.45 & & \end{array}$$

Then rule one again: both cells in $[0, 1]$. With squares and cubes of the letter in the table, the system is not linear, and on Paper 2 the calculator solves it — but it returns **every** root, and choosing among them is still yours.

A frequency table with unknown frequencies is the same question. A dice game recorded 14 scores: 1 occurred f times, 2 five times, 3 g times, and the mean score was 2.5:

$$\begin{aligned} f + 5 + g &= 14, \\ \frac{f + 10 + 3g}{14} &= 2.5 \implies g = 8, \\ f &= 1 \end{aligned}$$

Here rule one reads differently: frequencies are **whole numbers**, not below zero — and not below one, if the question says so.

The second fact can be about two variables. If X and Y are independent, $P(X < Y)$ is a sum over pairs of values:

$$P(X < Y) = \sum_{x < y} P(X = x) P(Y = y)$$

— list the pairs where the first is smaller; the ties $x = y$ are not in it.

Ranges. When the question gives no second fact, a second letter is *not* determined — and "*the range of possible values*" is what is asked. Use the sum to write every cell in one letter, then apply rule one to each cell. With m , $3m$, $1 - 4m$ from Part 1, $E(X) = m + 6m + 3 - 12m = 3 - 5m$, and as m runs over $0 \leq m \leq \frac{1}{4}$:

$$\frac{7}{4} \leq E(X) \leq 3$$

Task 4 — **May 2022 TZ2 Paper 1 Q10(a), 5 marks**

A biased four-sided die with faces labelled 1, 2, 3 and 4 is rolled and the result recorded. Let X be the result obtained when the die is rolled. The probability distribution for X is given in the following table where p and q are constants.

x	1	2	3	4
$P(X = x)$	p	0.3	q	0.1

For this probability distribution, it is known that $E(X) = 2$.

(a) Show that $p = 0.4$ and $q = 0.2$.

A "show that" is marked on the equations, so the equations are what you hand in — one from the sum, one from the mean — and then the values they give.

```
p, q = symbols('p q')

q4sum = ...      # the equation from the sum
of the probabilities
q4mean = ...     # the equation from  $E(X) = 2$ 
q4 = [...]      # [p, q]

X = Dist({1: p, 2: 0.3, 3: q, 4: 0.1})

verify_equation('4 sum', q4sum,
Eq(total_probability(X), 1), var=p)
verify_equation('4 mean', q4mean,
Eq(Expect(X), 2), var=p)
verify_letters('4', q4, [p, q], [X],
[Eq(Expect(X), 2)])
```

Task 5 — **November 2023 TZ1** **Paper 2 Q5, 5 marks**

The following table shows the probability distribution of a discrete random

variable X , where $a, k \in \mathbb{R}^+$.

x	1	2	3	4
$P(X = x)$	k	k^2	a	k^3

Given that $E(X) = 2.3$, find the value of a .

The cubic has three roots, and the calculator will show you all of them. The check solves for a and k together and knows which roots the table allows.

```
q5 = ...          # a

a, k = symbols('a k')
X = Dist({1: k, 2: k**2, 3: a, 4: k**3})

verify_letters('5', q5, a, [X],
[Eq(Expect(X), 2.3)])
```

Task 6 — **May 2021 TZ1 Paper 1 Q10, 16 marks**

A biased four-sided die, A , is rolled. Let X be the score obtained when die A is rolled. The probability distribution for X is given in the following table.

x	1	2	3	4
$P(X = x)$	p	p	p	$\frac{1}{2}p$

(a) Find the value of p .

(b) Hence, find the value of $E(X)$.

A second biased four-sided die, B, is rolled. Let Y be the score obtained when die B is rolled. The probability distribution for Y is given in the following table.

y	1	2	3	4
$P(Y = y)$	q	q	q	r

(c) (i) State the range of possible values of r . **(ii)** Hence, find the range of possible values of q .

(d) Hence, find the range of possible values for $E(Y)$.

Agnes and Barbara play a game using these dice. Agnes rolls die A once and Barbara rolls die B once. The probability that Agnes' score is less than Barbara's score is $\frac{1}{2}$.

(e) Find the value of $E(Y)$.

A Paper 1 question, so exact values: Rational(2, 7), and ranges as Interval(...). In (e) find q and r on the

way — the markscheme gives them a mark each.

```
q6a = ...          # p
q6b = ...          # E(X)
q6r = ...          # the range of r, as an
Interval
q6q = ...          # the range of q
q6d = ...          # the range of E(Y)
q6qr = [...]       # [q, r] in part (e)
q6e = ...          # E(Y) in part (e)

p, q, r = symbols('p q r')
X = Dist({1: p, 2: p, 3: p, 4: p/2}, 'X')
Y = Dist({1: q, 2: q, 3: q, 4: r}, 'Y')
game = [Eq(P(X < Y), Rational(1, 2))]      #
X and Y are independent

verify_letters('6a', q6a, p, [X])
verify_moment('6b', q6b, Expect(X), given=[],
var=p)
verify_table_range('6c(i)', q6r, r, [Y])
verify_table_range('6c(ii)', q6q, q, [Y])
verify_table_range('6d', q6d, Expect(Y), [Y])
verify_letters('6e q, r', q6qr, [q, r], [X,
Y], game)
verify_moment('6e', q6e, Expect(Y),
given=game, var=[p, q, r], tables=[X])
```

Theory: variance, and a variable built from X

Variance is the mean squared distance from the mean. Take

x	-1	0	2
$P(X = x)$	0.25	0.5	0.25

$E(X) = -0.25 + 0 + 0.5 = 0.25$, and the quick way to the variance is through $E(X^2)$ — the same weighted sum, with the values squared:

$$\begin{aligned} E(X^2) &= 1 \times 0.25 + 0 + 4 \times 0.25 \\ &= 1.25, \quad \text{Var}(X) \\ &= E(X^2) - E(X)^2 \\ &= 1.25 - 0.0625 \\ &= 1.1875 \end{aligned}$$

Two things go wrong here, and both are one line short: handing in $E(X^2) = 1.25$ as the variance, and subtracting $E(X) = 0.25$ instead of its square.

A variable built from X . A prize is $Y = 5 - 3X$ tokens. Then

$$E(Y) = 5 - 3E(X) = 4.25,$$

$$\text{Var}(Y) = (-3)^2$$

$$\text{Var}(X) = 9 \times 1.1875 = 10.6875$$

Adding moves the table, multiplying stretches it. The 5 shifts every value and the spread is untouched. The -3 stretches by 3 — and a variance is in squared units, so by 9. The sign disappears: $\text{Var}(-X) = \text{Var}(X)$.

When only E and Var are known. A city's daily temperature C in $^{\circ}\text{C}$ has $E(C) = 20$ and $\text{Var}(C) = 4$. In Fahrenheit, $F = 1.8C + 32$, so $E(F) = 68$ and $\text{Var}(F) = 1.8^2 \times 4 = 12.96$ — no table needed, because a linear change needs only those two numbers.

For a frequency table the variance divides by $\sum f$, like the mean: it describes the data, not an estimate from a sample of it.

Task 7 — **November 2025 TZ3**
Paper 1 Q7, 8 marks

The discrete random variable X has the following probability distribution:

x	1	2	3
$P(X = x)$	$0.6 - 2a$	$3a$	$0.4 - a$

(a) Find the range of possible values of a .

(b) In the case where $a = 0.2$, determine $\text{Var}(2 - X)$.

Part (a) is technique 1 — here the sum is one for every a , so only the cells decide. Part (b) is six marks for what the theory above does in two lines; the markscheme's first method starts with $\text{Var}(2 - X) = \text{Var}(X)$.

```
q7a = ...          # the range of a, as an
Interval
q7b = ...          # Var(2 - X)

a = symbols('a')
X = Dist({1: 0.6 - 2*a, 2: 3*a, 3: 0.4 - a})

verify_table_range('7a', q7a, a, [X])
verify_moment('7b', q7b, Var(2 - X),
given=Eq(a, 0.2), var=a)
```

Task 8 — May 2025 TZ2 Paper 2

Q8, 7 marks

The marks obtained by students in a class quiz are shown in the following table where $p, q \in \mathbb{Z}^+$.

marks	20	35	p
frequency	12	q	8

The mean and variance of the marks are 31 and 124 respectively. Find the value of p and the value of q .

Here one letter is a frequency and the other is a value — the table does not mind. The system has two solutions; the question's $\mathbb{Z}^+$ rules one of them out.

```
q8 = [...]          # [p, q]

p, q = symbols('p q', positive=True,
integer=True)
quiz = Freq({20: 12, 35: q, p: 8})

verify_letters('8', q8, [p, q], [quiz],
[Eq(Expect(quiz), 31), Eq(Var(quiz), 124)])
```

Task 9 ● — **May 2025 TZ1 Paper 2**
Q11(e)-(f), 7 marks

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. No runner completes the race in less than 2.25 hours, and no runner completes it in more than 7.5 hours.

Each runner's time is converted to a score which is calculated as $a - bt$, where t represents their time in hours, and $a, b > 0$. Consider the random variable P which represents the score of a runner. It is given that $E(P) = 100$ and the maximum possible score is 150.

(e) Use $E(T) = 4.723$ to determine the value of a and the value of b , giving your answers to the nearest integer.

(f) Given also that $\text{Var}(T) = 0.906$, find $\text{Var}(P)$.

T is continuous, and its density is D6's work — but $E(T)$ and $\text{Var}(T)$ are given, and a linear change needs nothing else. The check stands T in for two equally

likely values with that mean and that variance.

```
q9ab = [...]      # [a, b], to the nearest
integer
q9v = ...         # the variance of the score

a, b = symbols('a b', positive=True)
T = Moments(4.723, 0.906, 'T')
fastest = 2.25                                           #
no runner is faster
score = [Eq(Expect(a - b*T), 100), Eq(a -
b*fastest, 150)]

verify_letters('9e', q9ab, [a, b], [T],
score, whole=True)
verify_moment('9f', q9v, Var(a - b*T),
given=score, var=[a, b])
```

● Part 3. Paper 3: no last value, and the table as a polynomial

Theory: a first success

Roll a fair die until the first six. X , the number of rolls, can be $1, 2, 3, \dots$ with no largest value:

$$P(X = x) = \left(\frac{5}{6}\right)^{x-1} \frac{1}{6}$$

— $x - 1$ failures, then the success. The table has no end, but it is still a table, and the mean is still the weighted sum:

$$E(X) = \sum_{x=1}^{\infty} x \left(\frac{5}{6}\right)^{x-1} \frac{1}{6}$$

How to add it up. Start from a sum you know (A2): $1 + r + r^2 + r^3 + \dots = \frac{1}{1-r}$ for $|r| < 1$. Differentiate both sides with respect to r , term by term on the left:

$$1 + 2r + 3r^2 + 4r^3 + \dots = \frac{1}{(1-r)^2}$$

The left side is exactly the shape of the mean, with $r = \frac{5}{6}$:

$$\begin{aligned} E(X) &= \frac{1}{6} \left(1 + 2 \cdot \frac{5}{6} + 3 \left(\frac{5}{6} \right)^2 + \dots \right) \\ &= \frac{1}{6} \cdot \frac{1}{(1/6)^2} \\ &= 6 \end{aligned}$$

A six every six rolls, on average — which is what you expected, and now it is proved. Task 10 makes the same move with a letter instead of $\frac{1}{6}$. The variance takes a second derivative; the exam gives its formula, and for the die it is 30.

Two models with the same mean are compared by their variance. The smaller variance is the more *consistent* experience: the first success arrives closer to the same time, every time.

Task 10 ● — May 2024 TZ1 Paper 3 Q1(b)-(d), (g)(ii)-(h), 13 marks

In a new computer game, each time a player performs an action, there is a random chance that the action will be *boosted*. In the first model, the

probability that an action will be boosted is constant.

(b) Suppose the probability that an action will be boosted is p , where $0 < p < 1$. **(i)** Explain why the probability that the first boost occurs on the x th action is $p(1-p)^{x-1}$. Let X be the number of actions until the first boost occurs. **(ii)** Hence, write down an expression, using sigma notation, for $E(X)$ in terms of x and p .

(c) Part (c)(i) shows that $\sum_{n=1}^{\infty} nar^{n-1} = \frac{a}{(1-r)^2}$. **(ii)** Hence, show that $E(X) = \frac{1}{p}$.

It can be shown that $\text{Var}(X) = \frac{1-p}{p^2}$.

(d) Find $E(X)$ and $\text{Var}(X)$ when $p = 0.1$.

In the designer's second model, the probability of a boost starts at 0.2 and rises by 0.2 after every action that is not boosted. Let Y be the number of actions until the first boost occurs. Part (g)(i) finds the probability distribution of Y :

y	1	2	3	4	5
$P(Y = y)$	0.2	0.32	0.288	0.1536	0.0384

(g) (ii) Show that $E(Y) = 2.5104$. **(iii)** Find $\text{Var}(Y)$.

(h) (i) Use the expression given in (c)(ii) to find the value of p for which $E(X) = E(Y)$. **(ii)** Find $\text{Var}(X)$ for this value of p . **(iii)** Hence determine, with a reason, which model provides a more consistent experience for the player with respect to boosted actions.

(b)(i), (c)(ii) and (h)(iii) are words — they are in the solutions. The check for (b)(ii) evaluates your sum at several values of p ; write it with `Sum(..., (x, 1, oo))`.

```

p = symbols('p')

q10b = ...      # E(X) as a sum, in x and p
q10m = ...      # E(X) when p = 0.1
q10v = ...      # Var(X) when p = 0.1
q10ey = ...     # E(Y)
q10vy = ...     # Var(Y)
q10p = ...      # p with E(X) = E(Y)
q10vx = ...     # Var(X) for that p

X = Geo(p, 'X') #
the first boost, first model
Y = Dist({1: 0.2, 2: 0.32, 3: 0.288, 4:
0.1536, 5: 0.0384}, 'Y')
same_mean = Eq(Expect(X), Expect(Y))

verify_moment('10b(ii)', q10b, Expect(X))
verify_moment('10d E(X)', q10m,
Expect(Geo(0.1)))
verify_moment('10d Var(X)', q10v,
Var(Geo(0.1)))
verify_moment('10g(ii)', q10ey, Expect(Y))
verify_moment('10g(iii)', q10vy, Var(Y))
verify_letters('10h(i)', q10p, p, [X],
[same_mean])
verify_moment('10h(ii)', q10vx, Var(X),
given=same_mean, var=p)

```

Theory: a generating function

Write a table as a polynomial: the probability of x becomes the coefficient

of t^x . A spinner scoring 0, 1, 2 with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ has

$$G(t) = \frac{1}{2} + \frac{1}{3}t + \frac{1}{6}t^2$$

Nothing is lost — the table can be read back off the coefficients — and three facts come for free:

- $G(1) = 1$, because at $t = 1$ the polynomial is the sum of the table;
- $G'(1) = E(X)$: $G'(t) = \sum x P(X = x) t^{x-1}$, and at $t = 1$ that is the weighted sum. Here $G'(t) = \frac{1}{3} + \frac{1}{3}t$, so $E(X) = \frac{2}{3}$;
- **for independent X and Y , $G_{X+Y} = G_X G_Y$.** Multiplying out collects every pair of scores whose sum is the power of t .

Spin twice and add: $G(t)^2 = \frac{1}{4} + \frac{1}{3}t + \frac{5}{18}t^2 + \frac{1}{9}t^3 + \frac{1}{36}t^4$, and $E = 2 \times \frac{2}{3} = \frac{4}{3}$ — with $E(X + Y) = E(X) + E(Y)$ there is no need to multiply out at all.

The coefficients are built, not guessed. Each is a probability from the experiment — a tree, or combinations — and the markscheme

gives no marks for coefficients stated without that working.

Task 11 — **May 2025 TZ1 Paper 3**
Q1(a), (c)-(f), 22 marks

Two unbiased tetrahedral (four-sided) dice with faces labelled 1, 2, 3 and 4 are thrown and the scores recorded. The random variable M denotes the maximum of these two scores.

m	1	2	3	4
$P(M = m)$	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{7}{16}$

(a) Find $E(M)$.

The distribution of M is represented by $G(t) = \frac{1}{16}t + \frac{3}{16}t^2 + \frac{5}{16}t^3 + \frac{7}{16}t^4$.

(c) (i) Find $G'(t)$. **(ii)** Hence, show that $G'(1) = E(M)$.

A bag contains two red balls and three yellow balls. Two balls are selected at random without replacement from the bag. The random variable X denotes the total number of red balls selected, and $G_X(t) = \sum_{x=0}^2 P(X = x)t^x$.

(d) Show that $G_X(t) = \frac{3}{10} + \frac{3}{5}t + \frac{1}{10}t^2$, making it clear how the coefficients of $G_X(t)$ have been determined.

An unbiased coin and a biased coin are tossed. The probability of obtaining a tail on the biased coin is p . The random variable Y denotes the total number of tails obtained from tossing both coins, and $G_Y(t) = \sum_{y=0}^2 P(Y = y)t^y$.

(e) Given that the coefficient of t^2 in $G_Y(t)$ is $\frac{1}{3}$, find **(i)** the value of p ; **(ii)** an expression for $G_Y(t)$.

The random variable Z denotes the sum of X and Y , and $G_Z(t) = G_X(t)G_Y(t)$.

(f) For random variable Z , it can be shown that $G'_Z(1) = E(Z)$. Use this result to find $E(Z)$.

No check below holds a coefficient. M is built from the sixteen throws, X from the twenty ordered draws, Y as the sum of two coins.

```

q11a = ...      # E(M)
q11g = ...      # G'(t)
q11c = ...      # G'(1)
q11d = ...      # G_X(t)
q11p = ...      # p
q11y = ...      # G_Y(t)
q11z = ...      # E(Z)

throws = {(i, j): Rational(1, 16) for i in
range(1, 5) for j in range(1, 5)}
M = Dist(throws, 'M', rule=max)

balls = 'RRYYY'
draws = {(i, j): Rational(1, 20) for i in
range(5) for j in range(5) if i != j}
X = Dist(draws, 'X', rule=lambda pair:
(balls[pair[0]] == 'R') + (balls[pair[1]] ==
'R'))

p = symbols('p')
fair = Dist({0: Rational(1, 2), 1:
Rational(1, 2)}, 'fair')      # 1 is a tail
biased = Dist({0: 1 - p, 1: p}, 'biased')
Y = Dist(fair + biased, 'Y')
top = Eq(P(Y == 2), Rational(1, 3))

verify_moment('11a', q11a, Expect(M))
verify_derivative('11c(i)', q11g, Pgf(M, t),
var=t)
verify_moment("11c(ii) G'(1)", q11c,
Expect(M))
verify_pgf('11d', q11d, X, t)
verify_letters('11e(i)', q11p, p, [Y], [top])
verify_pgf('11e(ii)', q11y, Y, t, given=top,
unknowns=p)

```

```
verify_moment('llf', qllz, Expect(X + Y),  
given=top, var=p)
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
table	a letter in a table: the sum and the cells
mean	a mean, an expected number, a weighted sum
unknowns	two unknowns and one more fact about the distribution
spread	a variance, or a variable built from X
series	until the first success: values with no end
pgf	a generating function

1. $P(X = x) = cx$ for $x = 1, 2, 3, 4$. Find c .
2. A shop sells 0, 1, 2 or 3 cakes an hour with probabilities 0.1, 0.3, 0.4, 0.2. Find the expected number of cakes sold in an eight-hour day.

3. A table has cells $a, 0.2, b, 0.3$ for $x = 0, 1, 2, 3$ and $E(X) = 1.5$. Find a and b .
4. $E(X) = 3$ and $\text{Var}(X) = 2$. Find $\text{Var}(4 - 3X)$.
5. A basketball player shoots until she scores, each shot scoring with probability 0.35. Find the expected number of shots.
6. $G(t) = \frac{1}{4}(1 + t)^2$. Find $P(X = 1)$ and $E(X)$.
7. The frequencies 3, f , g , 2 of the scores 1, 2, 3, 4 total 15, and the mean score is 2.4. Find f and g .
8. A table has cells $0.5 - k, 2k, 0.5 - k$. State the range of possible values of k .
9. Each score in a data set is multiplied by 5 and then 3 is subtracted. The old standard deviation was 1.2. Find the new variance.
0. X takes the values $-2, 0, 3$ with probabilities 0.3, 0.5, 0.2. Find $E(X)$.
1. Two independent variables have $G_X(t) = 0.4 + 0.6t$ and $G_Y(t) = 0.7 + 0.3t$. Find the distribution of $X + Y$.

2. A machine fails each day with probability 0.02, independently. Find the variance of the number of days up to and including the first failure.

```
answers = {  
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',  
    7: '', 8: '', 9: '', 10: '', 11: '', 12:  
    '',  
}  
  
trigger_check(answers, {1: '0d4fc4a78d37', 2:  
'82523faea1b0', 3: '2041e4cdf428', 4:  
'909802b269de', 5: '9cda38ea3a2b', 6:  
'e10f417d699c', 7: '2041e4cdf428', 8:  
'0d4fc4a78d37', 9: '909802b269de', 10:  
'82523faea1b0', 11: 'e10f417d699c', 12:  
'9cda38ea3a2b'})
```

On the clock — May 2024 TZ1 Paper 1 Q1, 6 marks

Six marks, seven minutes. No calculator, no hints.

Claire rolls a six-sided die 16 times. The scores obtained are shown in the following frequency table.

score	1	2	3	4	5	6
frequency	p	q	4	2	0	3

It is given that the mean score is 3.

(a) Find the value of p and the value of q .

Each of Claire's scores is multiplied by 10 in order to determine the final score for a game she is playing.

(b) Write down the mean final score.

Attempt log

date	time	result

```
qt_pq = [...]      # [p, q]
qt_b = ...          # the mean final score

p, q = symbols('p q', integer=True,
nonnegative=True)
rolls = Freq({1: p, 2: q, 3: 4, 4: 2, 5: 0,
6: 3})
claire = [Eq(rolls.size, 16),
Eq(Expect(rolls), 3)]

verify_letters('timer (a)', qt_pq, [p, q],
[rolls], claire)
verify_moment('timer (b)', qt_b, Expect(10 *
rolls), given=claire, var=[p, q])
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a)(i) The cells add to one:

$$\begin{aligned}1.5a + 2a + 0.281 + a + 0.026 + 0 &= 1 \\ &\implies 4.5a \\ &= 0.693 \\ &\implies \boxed{a = 0.154}\end{aligned}$$

1 (a)(ii) The probabilities are 0.231, 0.308, 0.281, 0.154, 0.026: the largest belongs to $x = 2$, so the mode is $\boxed{2}$ visits.

1 (b)

$$\begin{aligned}E(X) &= 1(0.231) \\ &\quad + 2(0.308) \\ &\quad + 3(0.281) \\ &\quad + 4(0.154) \\ &\quad + 5(0.026) \\ &= 2.436 \\ &= \boxed{2.44}\end{aligned}$$

The markscheme prints 2.436 as exact: every cell is known to three decimal places, so the mean is too.

2 (a)

$$\begin{aligned}0.41 + (k - 0.28) + 0.46 + (0.29 - 2k^2) &= 1 \\ \implies 0.88 + k - 2k^2 & \\ = 1 & \\ \implies \boxed{2k^2 - k + 0.12 = 0}\end{aligned}$$

2 (b) $k = \frac{1 \pm \sqrt{1 - 0.96}}{4} = \frac{1 \pm 0.2}{4}$, so $k = 0.3$ or $k = 0.2$. But $k = 0.2$ makes $P(X = 1) = 0.2 - 0.28 = -0.08 < 0$, which is not a probability. $\boxed{k = 0.3}$ — and the sentence about $P(X = 1)$ is the R1.

2 (c) With $k = 0.3$ the table is 0.41, 0.02, 0.46, 0.11:

$$E(X) = 0 + 0.02 + 0.92 + 0.33 = \boxed{1.27}$$

3 (a)(i) 34 of the 40 rode at least once:
 $\frac{34}{40} = \boxed{0.85}$.

3 (a)(ii)

$$\frac{0(6) + 1(16) + 2(13) + 3(2) + 4(3)}{40} = \frac{60}{40} = \boxed{1.5}$$

3 (b) $1000 \times 1.5 = 1500$ rides expected, at 10 a run: $\boxed{150}$ runs.

4 From the sum: $p + 0.3 + q + 0.1 = 1$, so $\boxed{p + q = 0.6}$. From the mean: $p + 0.6 + 3q + 0.4 = 2$, so $\boxed{p + 3q = 1}$.

Subtracting, $2q = 0.4$: $\boxed{q = 0.2, p = 0.4}$. The markscheme gives the two equations two marks each, independently, and one for solving them.

5 Two equations:

$$\begin{aligned} k + k^2 + a + k^3 &= 1, & k + 2k^2 + 3a + 4k^3 \\ &= 2.3 \end{aligned}$$

Eliminate a ($a = 1 - k - k^2 - k^3$): $k^3 - k^2 - 2k + 0.7 = 0$, with roots $k = 0.315870\dots$, $-1.18538\dots$ and $1.86951\dots$. Only the first leaves every cell in $[0, 1]$ — the cell $P(X = 1) = k$ is negative for the second

and above one for the third — and it gives

$$\begin{aligned} a &= 1 - k - k^2 - k^3 \\ &= 0.552839 \dots \\ &= \boxed{0.553} \end{aligned}$$

The markscheme note: $a = 2.44587$ or $a = -10.8987$ from the other roots scores the method and loses the answer, and 0.55 loses it too.

6 (a) $3p + \frac{1}{2}p = 1$: $\boxed{p = \frac{2}{7}}$.

6 (b) $E(X) = p(1 + 2 + 3) + 4 \cdot \frac{1}{2}p = 8p = \boxed{\frac{16}{7}}$.

6 (c)(i) r is a probability: $\boxed{0 \leq r \leq 1}$.

6 (c)(ii) $3q = 1 - r$, so $q = \frac{1-r}{3}$, and as r runs from 1 to 0: $\boxed{0 \leq q \leq \frac{1}{3}}$.

6 (d) $E(Y) = q + 2q + 3q + 4r = 6q + 4(1 - 3q) = 4 - 6q$, and over $0 \leq q \leq \frac{1}{3}$: $\boxed{2 \leq E(Y) \leq 4}$.

6 (e) Agnes' score is less than Barbara's in the pairs (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4):

$$\begin{aligned} P(X < Y) &= p(2q + r) + p(q + r) + pr \\ &= 3p(q + r) \\ &= \frac{6}{7}(q + r) \\ &= \frac{1}{2} \end{aligned}$$

so $q + r = \frac{7}{12}$. With $3q + r = 1$: $2q = \frac{5}{12}$,

$$\boxed{q = \frac{5}{24}}, \boxed{r = \frac{3}{8}}, \text{ and}$$

$$E(Y) = 4 - 6 \cdot \frac{5}{24} = \boxed{\frac{11}{4}}$$

7 (a) Every cell at least zero: $0.6 - 2a \geq 0$, $3a \geq 0$, $0.4 - a \geq 0$. Together, $\boxed{0 \leq a \leq 0.3}$ — the middle condition is $a \leq 0.4$, and it is the first cell that binds. The sum is 1 for every a .

7 (b) $\text{Var}(2 - X) = (-1)^2 \text{Var}(X) = \text{Var}(X)$. With $a = 0.2$ the table is 0.2, 0.6, 0.2: $E(X) = 2$ by symmetry, $E(X^2) = 0.2 + 2.4 + 1.8 = 4.4$, and

$$\text{Var}(2 - X) = 4.4 - 2^2 = \boxed{0.4}$$

8 $\sum f = 20 + q$. Mean:

$$\begin{aligned}\frac{240 + 35q + 8p}{20 + q} &= 31 \\ &\implies 4q + 8p \\ &= 380 \\ &\implies q \\ &= 95 - 2p\end{aligned}$$

Variance, through the squared deviations from 31:

$$\frac{12(11)^2 + 16q + 8(p - 31)^2}{20 + q} = 124$$

Substituting $q = 95 - 2p$ gives a quadratic in p with roots $p = 45$ and $p = -10$. The question says $p \in \mathbb{Z}^+$: $\boxed{p = 45, q = 5}$. The markscheme note: giving both pairs scores A1A0.

9 (e) $E(P) = a - bE(T)$, so $a - 4.723b = 100$. The best score belongs to the fastest time, $t = 2.25$: $a - 2.25b = 150$. Subtracting, $2.473b = 50$:

$$b = 20.2183 \dots,$$

$$a = 195.491 \dots$$

$$\boxed{a = 195, b = 20}$$

9 (f) $\text{Var}(P) = (-b)^2 \text{Var}(T) = 20.2183 \dots^2 \times 0.906 = 370.356 \dots = \boxed{370}$

With $b = 20$ carried from (e) the answer is 362 — the markscheme accepts anything from 362 to 370. The check accepts it too, with a remark.

10 (b)(i) The first boost on the x th action means $x - 1$ actions not boosted, each with probability $1 - p$, then one boosted: $p(1 - p)^{x-1}$.

10 (b)(ii)
$$E(X) = \sum_{x=1}^{\infty} x p (1 - p)^{x-1}$$

10 (c)(ii) This is $\sum n a r^{n-1}$ with $a = p$, $r = 1 - p$:

$$E(X) = \frac{p}{(1 - (1 - p))^2} = \frac{p}{p^2} = \frac{1}{p}$$

10 (d) $E(X) = \boxed{10}$, $\text{Var}(X) = \frac{0.9}{0.01} = \boxed{90}$.

$$\mathbf{10} \quad \mathbf{(g)(ii)} \quad 0.2 + 2(0.32) + 3(0.288) + 4(0.1536) + 5(0.0384) = 2.5104 \checkmark$$

$$\mathbf{10} \quad \mathbf{(g)(iii)} \quad E(Y^2) = 0.2 + 4(0.32) + 9(0.288) + 16(0.1536) + 25(0.0384) = 7.4896, \text{ so}$$

$$\text{Var}(Y) = 7.4896 - 2.5104^2$$

$$= 1.18749 \dots$$

$$= \boxed{1.19}$$

$$\mathbf{10} \quad \mathbf{(h)(i)} \quad \frac{1}{p} = 2.5104: \quad p = 0.398342 \dots =$$

$$\boxed{0.398}$$

$$\mathbf{10} \quad \mathbf{(h)(ii)} \quad \text{Var}(X) = \frac{1 - 0.398342 \dots}{0.398342 \dots^2} =$$

$$3.79170 \dots = \boxed{3.79}$$

(The markscheme accepts 3.80, from $p = 0.398$.)

10 (h)(iii) The means are equal and $\text{Var}(Y) = 1.19 < 3.79 = \text{Var}(X)$: the **second model** is more consistent.

$$\mathbf{11} \quad \mathbf{(a)} \quad E(M) = \frac{1 + 6 + 15 + 28}{16} = \frac{50}{16} =$$

$$\boxed{3.125}$$

$$\mathbf{11 (c)(i)} \quad G'(t) = \boxed{\frac{1}{16} + \frac{3}{8}t + \frac{15}{16}t^2 + \frac{7}{4}t^3}$$

$$\mathbf{11 (c)(ii)} \quad G'(1) = \frac{1 + 6 + 15 + 28}{16} = 3.125 =$$

$E(M) \checkmark$

11 (d) Two draws without replacement, from 2 red and 3 yellow:

$$P(X = 0) = \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{10},$$

$$P(X = 1) = \frac{2}{5} \cdot \frac{3}{4} + \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{5},$$

$$P(X = 2) = \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}$$

and these are the coefficients:

$$\boxed{G_X(t) = \frac{3}{10} + \frac{3}{5}t + \frac{1}{10}t^2}.$$

11 (e)(i) Two tails needs a tail on both:

$$\frac{1}{2}p = \frac{1}{3}, \text{ so } \boxed{p = \frac{2}{3}}.$$

$$\mathbf{11 (e)(ii)} \quad P(Y = 0) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}, \quad P(Y = 1) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{2}:$$

$$\boxed{G_Y(t) = \frac{1}{6} + \frac{1}{2}t + \frac{1}{3}t^2}$$

With $p = \frac{1}{3}$ instead, the coefficients come out reversed — the heads, not the tails.

11 (f) $G'_Z(1) = G'_X(1) + G'_Y(1)$, because $G_X(1) = G_Y(1) = 1$:

$$\begin{aligned} E(Z) &= \left(\frac{3}{5} + \frac{2}{10}\right) + \left(\frac{1}{2} + \frac{2}{3}\right) \\ &= \frac{4}{5} + \frac{7}{6} \\ &= \frac{59}{30} \\ &= \boxed{1.97} \end{aligned}$$

Timer

(a) Sixteen rolls: $p + q + 9 = 16$, so $p + q = 7$. Mean three: $p + 2q + 12 + 8 + 18 = 48$, so $p + 2q = 10$. Hence $\boxed{q = 3, p = 4}$.

(b) Every score multiplied by 10 multiplies the mean by 10: $\boxed{30}$.